Smoking and Skin Aging

Cigarette smoking exacerbates photoaging, particularly in women, with a direct correlation between the number of pack-years smoked and the severity of wrinkling and grayish discoloration. Histologic analysis of "smoker's skin" reveals elastic fiber thickening and fragmentation, similar to that found in sun-damaged skin. However, whereas solar elastosis is restricted to the papillary dermis, elastic fiber changes in smoker's skin also occur in the reticular dermis.

This dermal elastosis has been suggested to result from increased elastase activity in neutrophils, chronic dermal ischemia, and the pro-oxidant effects of cigarette smoke compounded by decreased levels of vitamin A, which reduce the capacity to quench free oxygen radicals and increase DNA damage. Smoking also has been associated with decreased stratum corneum water content and accelerated hydroxylation of estradiol, leading to decreased estrogens in the skin that may in turn contribute to dryness and atrophy. Smokers display poor wound healing capacity and increased incidence of skin cancers, as well as increased severity of photoaging-like changes compared with nonsmokers with otherwise similar risk factors, consistent with the possibility that mutagens present in cigarette smoke directly affect cells in the dermis and epidermis.

Relevance to Skin Disease in the Elderly

Closely associated with aging is increased vulnerability to disease and injury. Disorders of the skin are common and often bothersome in the elderly, with some occurring predominantly in this age group. Such disorders often appear to be the consequence of age-associated intrinsic losses of cutaneous cellular function. However, many dermatoses observed more commonly in the elderly reflect the higher prevalence of systemic diseases—such as diabetes, vascular insufficiency, and various neurologic syndromes—that compound physiologic changes in the skin itself. The increased prevalence of some disorders may also reflect reduced local skin care due to immobility or neurologic impairment rather than intrinsic skin changes. Additionally, subtle age-associated changes in immune status may contribute, analogous to the increased prevalence and severity of skin disease in patients with acquired immunodeficiency syndrome.

Reduced tolerance to systemically administered drugs is well documented in the elderly due to decrements in lean body mass, metabolic activity, and renal excretion of active ingredients. Comparable data for topically applied medications are lacking, but it is plausible that retarded dermal clearance of absorbed material, reduced dermal mass and cellularity, and possibly altered metabolic capacity may render aged skin more susceptible to both beneficial and adverse effects of topical medications—or at least alter the optimal therapeutic schedule. In the case of glucocorticoid preparations, relative vascular unresponsiveness may render blanching of erythema an unreliable indicator of other effects in aged skin.

Selected common cutaneous disorders in the elderly are discussed briefly in the following sections, with special emphasis on their pathogenesis and clinical presentation in this population. Most of these entities are covered more comprehensively in other chapters.

Tumors

Skin Cancer (See Chaps. 113 to 115)

The age-specific incidence of skin cancer, including melanoma, increases exponentially with age, presumably due in part to cumulative exposure to carcinogens over a lifetime interspersed with cell division, with the attendant risk of mutation. There are also well-documented age-associated decreases in DNA repair capacity and immunosurveillance, as well as subtle loss of proliferative homeostasis.

The major etiologic factor for skin cancer is UV irradiation. Habitual sun exposure of fair-skinned individuals induces squamous cell carcinoma as well as actinic keratoses, a premalignant lesion that may evolve into squamous cell carcinoma in a direct dose-response relationship. In contrast, the risk of basal cell carcinomas (BCCs) and particularly malignant melanoma is related not only to total sun exposure but also to intense intermittent sun exposure. The mechanism underlying these different epidemiologic patterns is unknown, although inducible protective responses and relative resistance of the responsible cell types to apoptosis may contribute.

The elderly, particularly men, present with melanomas (see Chap.